

Project output:

Part one:

Search:

The screenshot shows a REST client interface with the following details:

- Request:** GET `http://localhost:5050/search/distributed systems`
- Auth Type:** Inherit a...
- Response Status:** 200 OK • 110 ms • 341 B
- Response Body (JSON):**

```
1  [
2    {
3      "id": 1,
4      "title": "How to get a good grade in DOS in 40 minutes a day"
5    },
6    {
7      "id": 2,
8      "title": "RPCs for Noobs"
9    }
10 ]
```

Information using id:

GET http://localhost:5050/info/4 Send

Docs Params **Authorization** Headers 7 Body Scripts Settings Cookies

Auth Type
Inherit a...

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

No Auth
This request does not use any authorization. ⓘ

Body Cookies Headers 7 Test Results 200 OK • 40 ms • 306 B • Save Response ...

{ } JSON Preview Visualize

```
1 {  
2   "title": "Cooking for the Impatient Undergrad",  
3   "quantity": 4,  
4   "price": 20  
5 }
```

Purchase:

HTTP y/h > New Request Save Share

POST http://localhost:5050/purchase/4 Send

Docs Params **Authorization** Headers 8 Body Scripts Settings Cookies

Auth Type
Inherit a...

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

No Auth
This request does not use any authorization. ⓘ

Body Cookies Headers 7 Test Results 200 OK • 109 ms • 264 B • Save Response ...

{ } JSON Preview Visualize

```
1 {  
2   "msg": "purchase successful"  
3 }
```

output:

```

=> [catalog-service-1] resolving provenance for metadata file 0.0s
[+] Running 7/7
✓ bazar-node-catalog-service Built 0.0s
✓ bazar-node-order-service Built 0.0s
✓ bazar-node-frontend-service Built 0.0s
✓ Network bazar-node_default Created 0.0s
✓ Container bazar-node-catalog-service-1 Created 0.0s
✓ Container bazar-node-order-service-1 Created 0.0s
✓ Container bazar-node-frontend-service-1 Created 0.0s
Attaching to catalog-service-1, frontend-service-1, order-service-1
frontend-service-1 | Frontend service running on port 5000
frontend-service-1 | [SEARCH] topic = distributed systems
frontend-service-1 | [SEARCH] success
frontend-service-1 | [INFO] request for book id = 4
frontend-service-1 | [INFO] success id = 4
frontend-service-1 | [PURCHASE] start id = 4
frontend-service-1 | [PURCHASE] success id = 4

```

Part 2:

```

=> [catalog-service-2] resolving provenance for metadata file 0.0s
[+] Running 11/11
✓ bazaryh-order-service-2 Built 0.0s
✓ bazaryh-order-service-1 Built 0.0s
✓ bazaryh-frontend-service Built 0.0s
✓ bazaryh-catalog-service-1 Built 0.0s
✓ bazaryh-catalog-service-2 Built 0.0s
✓ Network bazaryh_default Created 0.0s
✓ Container bazaryh-catalog-service-2-1 Created 0.1s
✓ Container bazaryh-catalog-service-1-1 Created 0.1s
✓ Container bazaryh-order-service-2-1 Created 0.3s
✓ Container bazaryh-order-service-1-1 Created 0.3s
✓ Container bazaryh-frontend-service-1 Created 0.0s
Attaching to catalog-service-1-1, catalog-service-2-1, frontend-service-1, order-service-1-1, order-service-2-1
catalog-service-1-1 | Catalog Replica 1 running on 5001
catalog-service-2-1 | Catalog Replica 2 running on 5001
order-service-1-1 | Order Replica 1 running on 5002
order-service-2-1 | Order Replica 2 running on 5002
frontend-service-1 | Frontend service running on port 5000

```

[View in Docker Desktop](#)
[View Config](#)
[Enable Watch](#)

Search

REST client interface showing a GET request to `http://localhost:5050/search/distributed systems`. The response is a 200 OK status with a JSON body containing two items:

```
[{"id": 1, "title": "How to get a good grade in DOS in 40 minutes a day"}, {"id": 2, "title": "RPCs for Noobs"}]
```

Load balancing (round robin):

```
catalog-service-1-1 | [CATALOG 1] search("distributed systems")
catalog-service-2-1 | [CATALOG 2] search("distributed systems")
catalog-service-1-1 | [CATALOG 1] search("distributed systems")
```

Information using id (cash miss):

HTTP yh > New Request

Save Share

GET http://localhost:5050/info/4 Send

Docs Params Authorization Headers 7 Body Scripts Settings

Auth Type
Inherit a...

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

—

No Auth

This request does not use any authorization. ⓘ

Body Cookies Headers 7 Test Results ⓘ

200 OK • 51 ms • 342 B • ⓘ Save Response ...

{ } JSON Preview Visualize

```
1 {
2   "id": 4,
3   "title": "Cooking for the Impatient Undergrad",
4   "topic": "undergraduate school",
5   "stock": 1,
6   "price": 20
7 }
```

Information using id (cash hit):

GET http://localhost:5050/info/4 Send

Docs Params **Authorization** Headers 7 Body Scripts Settings Cookies

Auth Type
Inherit a...

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

No Auth
This request does not use any authorization. ⓘ

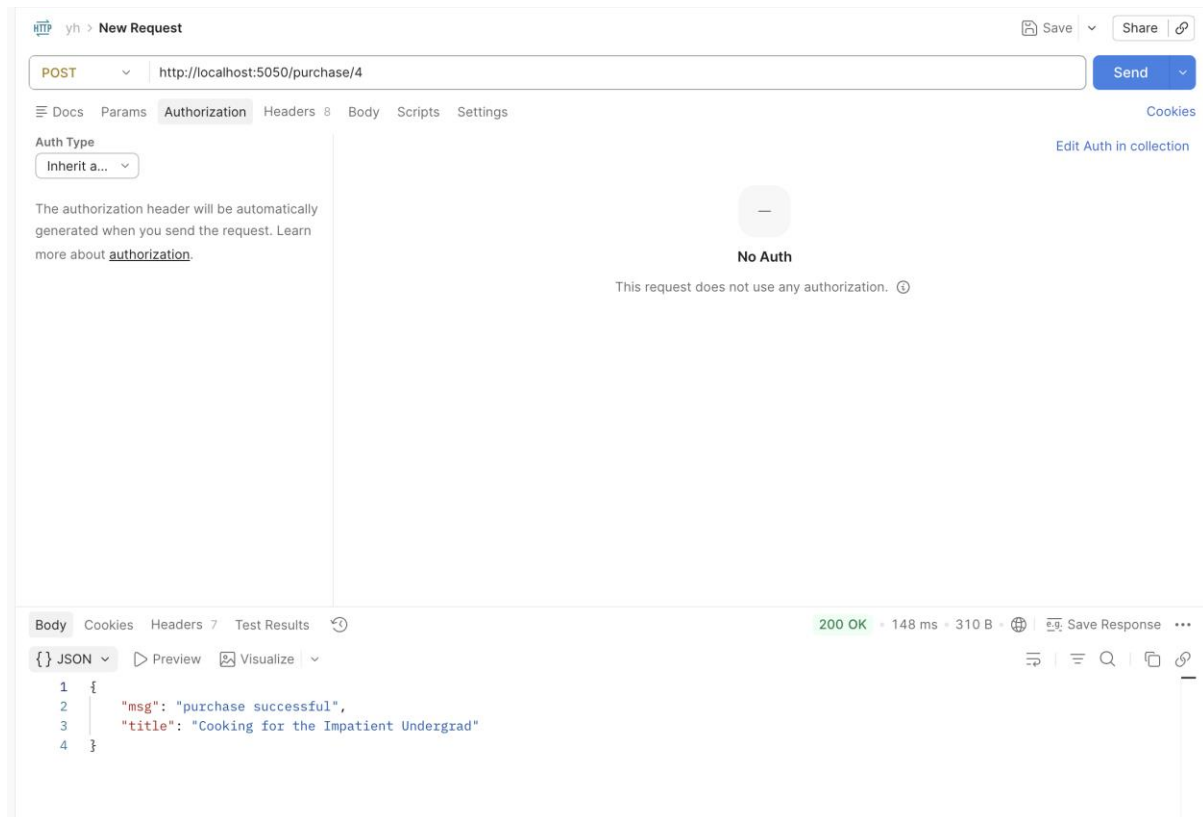
Body Cookies Headers 7 Test Results 200 OK • 5 ms • 342 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": 4,
3   "title": "Cooking for the Impatient Undergrad",
4   "topic": "undergraduate school",
5   "stock": 1,
6   "price": 20
7 }
```

```
catalog-service-2-1 | [CATALOG 2] info(4)
frontend-service-1 | CACHE HIT
frontend-service-1 | CACHE HIT
```

Purchase:



Cache Invalidation Overhead

First get after purchase (cash miss)

yt yh > New Request Save Share

GET http://localhost:5050/info/4 Send

Docs Params **Authorization** Headers 7 Body Scripts Settings Cookies

Auth Type
Inherit a...

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

No Auth
This request does not use any authorization. ⓘ

Body Cookies Headers 7 Test Results 200 OK · 65 ms · 342 B · Save Response

{ } JSON Preview Visualize

```
1 {  
2   "id": 4,  
3   "title": "Cooking for the Impatient Undergrad",  
4   "topic": "undergraduate school",  
5   "stock": 0,  
6   "price": 20  
7 }
```

Cash hit:

yt yh > New Request Save Share

GET http://localhost:5050/info/4 Send

Docs Params **Authorization** Headers 7 Body Scripts Settings Cookies

Auth Type
Inherit a...

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).

No Auth
This request does not use any authorization. ⓘ

Body Cookies Headers 7 Test Results 200 OK · 4 ms · 342 B · Save Response

{ } JSON Preview Visualize

```
1 {  
2   "id": 4,  
3   "title": "Cooking for the Impatient Undergrad",  
4   "topic": "undergraduate school",  
5   "stock": 0,  
6   "price": 20  
7 }
```


When out of stock

The screenshot shows a REST client interface with a POST request to `http://localhost:5050/purchase/4`. The 'Authorization' tab is selected, showing 'Auth Type' set to 'Inherit a...'. A message states: 'The authorization header will be automatically generated when you send the request. Learn more about [authorization](#).' The response area shows a 'No Auth' status with the message 'This request does not use any authorization.' The bottom panel shows the response body as JSON: `{ "msg": "out of stock" }`. The status bar indicates a '400 Bad Request' with a response time of 54 ms and a size of 266 B.

```
1 id,title,topic,stock,price
2 1,"How to get a good grade in DOS in 40 minutes a day","distributed systems",1,50
3 2,"RPCs for Noobs","distributed systems",0,40
4 3,"Xen and the Art of Surviving Undergraduate School","undergraduate school",0,30
5 4,"Cooking for the Impatient Undergrad","undergraduate school",0,20
6 5,"How to finish Project 3 on time","general",10,30
7 6,"Why theory classes are so hard","general",8,25
8 7,"Spring in the Pioneer Valley","general",12,40
9
```